

MULTI-DIMENSIONAL ANALYSIS ON COLLABORATIVE ANNOTATIONS

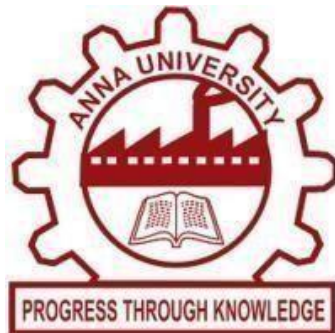
A THESIS

Submitted By

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in partial fulfillment for the award of the degree of

**MASTER OF TECHNOLOGY
in
INFORMATION TECHNOLOGY**



**DEPARTMENT OF INFORMATION TECHNOLOGY
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APRIL / MAY 2025

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BONAFIDE CERTIFICATE

Certified that this Report titled “**MULTI-DIMENSIONAL ANALYSIS ON COLLABORATIVE ANNOTATIONS**” is the Bonafide work of **JANANI GOPALAKRISHNAN (2023611026)** who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Cricket analytics is advancing with the integration of artificial intelligence (AI) and natural language processing (NLP) to offer real-time insights, match predictions, and contextual information. Conventional approaches to analyzing cricket data, such as manual extraction from sources like Cricsheet, are time-consuming and inefficient. This project introduces an AI-powered cricket chatbot to address these limitations, providing instant, accurate responses and voice-based interactions to enhance the experience for cricket enthusiasts and analysts alike.

The developed system comprises modules for context-based retrieval, match insights, predictive analysis, voice modeling, visualization, and a web interface. It employs NLP techniques, including regular expressions and Named Entity Recognition (NER), to extract entities like teams and dates, while logistic regression predicts match outcomes. Pandas and MongoDB handle statistical aggregation and data storage, and speech-to-text (STT) with Google Speech Recognition API and text-to-speech (TTS) with pyttsx3 enable seamless voice communication. These components collaboratively deliver dynamic, multilingual cricket insights in English and Tamil.

The chatbot was built using Visual Studio Code as the development platform, MongoDB for efficient data management, and a web interface crafted with HTML, CSS, and JavaScript, enhanced by Chart.js for visualizations. The backend leverages FastAPI for API development, with Requests library facilitating integration of real-time data from Cricsheet, CricketData, Google News, and Wikipedia APIs. This robust technological stack ensures a scalable, user-friendly platform for AI-driven cricket analysis.

ACKNOWLEDGEMENT

I render my sincere thanks to my project guide, **Dr. P. ANANDHA KUMAR**, Professor, Department of Information Technology, Anna University, MIT Campus for his valuable guidance, encouragement and moral support during this project. His systematic approach guided me to shape my project within the stipulated time.

I wish to express my sincere thanks to **Dr. M.R. SUMALATHA**, Professor and Head Department of Information Technology, Anna University, MIT Campus for providing all the facilities that were needed for proper execution of this project.

I wish to extend my sincere thanks and gratitude to the panel members, **Dr. M. R. SUMALATHA**, Professor, **Dr. R. GEETHA RAMANI**, Professor, and **Dr. P. ANANDHA KUMAR**, Professor, for their valuable suggestions and encouragement during the reviews.

I express my sincere thanks to **Dr. RAVICHANDRAN KANDASWAMY**, Professor and Dean, Anna University, MIT Campus, for providing me an opportunity to carry out the project work with all the facilities.

I thank my faculty advisor, all the teaching and non-teaching staff members of Department of Information Technology, for their support in all aspects. This project would not be at this stage without the co-operation extended by the almighty, family and friends. I am immensely grateful to all the people who helped me directly and indirectly in the successful completion of this project.

JANANI GOPALAKRISHNAN

(2023611026)

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LIST OF ABBREVIATIONS

ABBREVIATIONS	EXPANSION
AI	Artificial Intelligence
NLP	Natural Language Processing
NER	Named Entity Recognition
STT	Speech-to-Text
TTS	Text-to-Speech
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
API	Application Programming Interface

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The domain of artificial intelligence (AI) and natural language processing (NLP) has witnessed significant advancements, transforming how users interact with information systems. AI-powered chatbots, in particular, have emerged as a pivotal technology, enabling seamless, context-aware, and interactive communication across various sectors, including sports analytics. The sports industry, with its vast data repositories and enthusiastic fanbase, presents a unique opportunity for AI applications. Cricket, a globally celebrated sport with a massive following, generates extensive data through match statistics, player performances, and real-time news. Leveraging this data, AI systems can provide personalized, multilingual, and data-driven insights, enhancing fan engagement and accessibility.

AI chatbots in the sports domain integrate diverse data sources, such as historical datasets, real-time APIs, and unstructured content from news and encyclopedic platforms, to deliver comprehensive responses. These systems employ NLP techniques to process user queries, ranging from simple factual questions to complex analytical requests, and utilize machine learning models for predictive analytics. Furthermore, the incorporation of voice-based interfaces and multilingual support broadens accessibility, catering to diverse linguistic communities. Visualization tools complement these systems by presenting data in intuitive formats, such as graphs and charts, making insights more digestible for users.

The integration of collaborative annotations—where multiple data sources are aggregated and analyzed—enhances the accuracy and relevance of AI responses. In the context of cricket, this approach enables systems to provide context-based match summaries, player statistics, and predictive outcomes, addressing the needs of fans, analysts, and enthusiasts. Additionally, advancements in voice modeling allow for natural and interactive user experiences, supporting both text and voice inputs/outputs in multiple languages. This domain's interdisciplinary nature,

combining AI, data science, and user interface design, underscores its potential to redefine how sports-related information is accessed and consumed, fostering a more inclusive and engaging experience for global audiences.

1.2 PROBLEM STATEMENT

Cricket fans often seek real-time, context-aware, and multilingual information about matches, player statistics, and predictive insights. Existing platforms may lack seamless integration of diverse data sources, multilingual support, or interactive voice-based interfaces. Additionally, there is a need for accurate predictive models and visualizations to enhance user engagement. The absence of a unified system that combines collaborative annotations, voice-based interaction, and predictive analytics limits the accessibility and depth of cricket-related information. This project addresses these challenges by developing an AI cricket chatbot that provides a holistic, multilingual, and interactive experience.

1.3 OBJECTIVES

The main objectives of this project are:

- To develop an AI-driven cricket chatbot capable of understanding and responding to user queries contextually.
- To integrate data from multiple sources including Cricsheet, Wikipedia, CricketData, and Google News.
- To support multi-modal interaction through both voice and text, with input and output in English and Tamil.
- To summarize matches and provide real-time insights.
- To enhance user experience through data visualization.
- To analyze collaborative annotations and improve the chatbot's sensitivity to contextual nuances.

1.4 ABOUT THE PROJECT

The project, titled "Multi-dimensional Analysis on Collaborative Annotations," develops an AI cricket chatbot with the following key components:

a. Data Integration:

The chatbot utilizes multiple data sources, including the Cricsheet dataset for historical match data, Wikipedia for contextual cricket information, CricketData for structured statistics, and Google News for real-time updates, to provide accurate and comprehensive responses.

b. Query Processing:

The system supports context-based query answering, enabling users to ask about match summaries, player statistics, or other cricket-related information, with responses tailored to the query's context.

c. Multilingual Interaction:

The chatbot accepts both text and voice input and provides outputs in English and Tamil, leveraging voice modeling to enhance accessibility for diverse users.

d. Predictive Analytics:

Using logistic regression, the system performs Predictive Analytics to forecast match outcomes and player performance, offering valuable insights for fans and analysts.

e. Visualization:

The chatbot generates visualizations, such as graphs and charts, to present match statistics and predictive trends in an intuitive and user-friendly format.

This project emphasizes collaborative annotations by aggregating and analyzing data from multiple sources to deliver relevant and accurate responses, creating an interactive and engaging platform for cricket enthusiasts.

1.5 ORGANIZATION OF THE REPORT

This report is organized to provide a comprehensive account of the project, structured into six chapters, followed by references. Chapter 1: Introduction outlines the domain, problem statement, objectives, project details, and report structure. Chapter 2: Literature Survey reviews related and existing works in AI chatbots, sports

analytics, and cricket data systems, highlighting challenges in the field. Chapter 3: System Design describes the system architecture and provides a detailed dataset description, covering the integration of Cricsheet, Wikipedia, CricketData, and Google News. Chapter 4: System Development explains the development process, detailing the implementation steps, tools, and techniques used to build the chatbot. Chapter 5: Results and Discussion presents the project outcomes, including performance evaluation, visualizations, and analysis of the system's effectiveness. Chapter 6: Conclusion and Future Work summarizes the project findings, contributions, and potential enhancements for future development. The report concludes with a References section, listing all cited sources.

1.6 SUMMARY

This chapter introduced the domain of AI and NLP in sports analytics, focusing on the potential of cricket chatbots to enhance fan engagement. The problem statement highlighted the need for a unified, interactive platform for cricket information, while the objectives outlined the project's goals. The chapter also provided an overview of the project, structured as key components, and detailed the organization of the report, setting the stage for subsequent chapters that will delve into the technical and analytical aspects of the system.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive review of related works, focusing on existing systems, datasets, methodologies, pros, cons, and challenges in the domains of chatbots.

2.1 CRICKET ANALYTICS AND CHATBOTS

Several studies have explored the application of AI and data analytics in cricket, focusing on predictive modeling, score prediction, and conversational systems. Dey et al. (2012) [16] employed cluster analysis on Indian Premier League (IPL) data to identify patterns in team performance, laying the groundwork for data-driven cricket analytics. Similarly, Sankaranarayanan et al.

(2014) [45] developed a data mining approach for one-day international (ODI) cricket simulation and prediction, using historical data to forecast match outcomes. Singh et al. (2015) [40] proposed a data mining framework for score and winning prediction in cricket, emphasizing the role of statistical models in outcome prediction. Shah (2017) [49] focused on predicting match outcomes using the Duckworth-Lewis par score, highlighting the importance of real-time data in dynamic predictions.

Lokhande (2018) [28] introduced a system for live cricket score prediction and winning probability estimation, integrating real-time data feeds. Modekurti (2020) [34] analyzed target score setting in T20 cricket, using statistical models to optimize batting strategies. Suresh et al. (2021) [50] designed a chatbot for IPL first-inning score prediction, combining conversational AI with predictive analytics. Gupta et al.

(2017) [22] explored fantasy league team prediction, leveraging player statistics for team composition. More recently, Tatawat and Ghosh (2024) [51] developed CricGPT, a GPT-aided question-answering system for cricket, which processes queries using large language models (LLMs) and cricket-specific datasets,

demonstrating the potential of AI-driven conversational systems in the sport. These works collectively underscore the growing use of AI in cricket for predictive analytics and user interaction, though few integrate multilingual voice interfaces or collaborative annotations from diverse data sources, as proposed in this project.

2.2 MEDICAL CHATBOTS

Medical chatbots have gained prominence for their ability to provide health-related information and emotional support, offering insights into conversational AI design. Denecke et al. (2020) [18] developed SERMO, a mental health chatbot for regulating emotions, which underwent usability testing to ensure effective user interaction. Polignano et al. (2020) [42] introduced HealthAssistantBot, a personal health assistant for Italian-speaking users, focusing on natural language understanding and context-aware responses.

These systems highlight the importance of domain-specific knowledge and user trust in chatbot interactions. While medical chatbots prioritize sensitive data handling and emotional intelligence, their approaches to NLP and user engagement are relevant to the cricket chatbot's context-based query processing and multilingual support. However, medical chatbots typically do not incorporate predictive analytics or visualizations, which are central to the current project.

2.3 EDUCATIONAL AND GENERAL -PURPOSE CHATBOTS

Educational and general-purpose chatbots provide a broader perspective on conversational AI, particularly in terms of usability and scalability. Abu Shawar and Atwell (2007, 2014) [3, 4] investigated the utility of chatbots, questioning their practical applications and highlighting early challenges in natural language understanding. Følstad and Skjuve (2019) [19] examined chatbots in customer service, focusing on user experience and motivation, which informs the design of user-friendly interfaces for the cricket chatbot.

Chen et al. (2023) [14] explored AI student assistants in classrooms, designing chatbots to support student success through personalized responses. AlAfnan et al.

(2023) [7] evaluated ChatGPT as an educational tool, identifying opportunities and challenges in communication and writing courses. Baidoo-Anu and Owusu Ansah (2023) [11] discussed the benefits of ChatGPT in education, emphasizing its role in promoting teaching and learning. Kohnke (2023) [28] proposed a pedagogical chatbot as a supplemental language learning tool, highlighting the importance of multilingual capabilities.

Alsanousi et al. (2023) [8] analyzed user reviews of AI-enabled learning apps, identifying usability issues that are relevant to the cricket chatbot's design. Belal et al. (2023) [12] leveraged ChatGPT for sentiment analysis in text annotation, demonstrating the versatility of LLMs in processing unstructured data.

Park et al. (2023) [40] studied prompt engineering for generative AI, offering techniques to improve LLM performance, which is applicable to the cricket chatbot's query processing.

2.4 OTHER DOMAIN OF CHATBOTS

AI applications in other domains, such as sports and geospatial science, offer additional context for the project. Nadikattu (2020) [35] explored AI implementations in sports, including performance analysis and fan engagement, which align with the cricket chatbot's objectives. Fujii et al. (2021) [20] analyzed AI strategies in football, using reinforcement learning to model game tactics, providing a framework for predictive modeling in sports.

Hammes et al. (2022) [24] reviewed AI in elite sports, discussing success stories and challenges in integrating AI with real-world data. In geospatial science, Wang et al. (2024) [52] conducted a systematic review of LLMs and generative AI, highlighting their applications in data analysis and visualization. Hadid et al. (2024) [23] examined generative AI in geoscience, identifying trends and challenges in data integration. Cahyana et al. (2024) [13] investigated ChatGPT's application in soil science research, focusing on user perceptions and data processing capabilities. These works demonstrate the versatility of AI across domains, particularly in data

integration and user interaction, but they do not address cricket-specific requirements or multilingual voice interfaces.

2.5 CHALLENGES IN EXISTING SYSTEM

The development of chatbots in cricket and medical domains faces several challenges, as identified in the literature:

- a. **Data Integration and Quality:** Diverse data sources, such as Cricsheet, Wikipedia, Google News for cricket, or PubMed and forums for medical chatbots, often have inconsistent formats, missing data, or varying reliability, complicating integration. Real-time APIs like Google News may introduce noise, reducing response accuracy [27, 36].
- b. **Multilingual Support:** Providing robust support for multiple languages (e.g., English and Tamil for cricket chatbots) requires extensive training data and advanced NLP models, which many systems lack, limiting accessibility for diverse users [27, 34, 43].
- c. **Accuracy and Hallucination:** Advanced large language models (LLMs) like GPT-4 may generate inaccurate or fabricated responses (hallucination), particularly critical in medical contexts. Cricket chatbots also struggle with unpredictable match outcomes, affecting prediction reliability [36, 41].
- d. **Voice Interaction:** Implementing accurate speech recognition and synthesis for voice-based input/output, especially in multilingual settings, is complex. Most existing systems are text-based, limiting accessibility for users preferring voice interaction [27, 41, 26].
- e. **User Engagement and Experience:** Maintaining natural, context-aware responses and intuitive interfaces is challenging. Many chatbots struggle with complex queries or sustaining long conversations, leading to reduced user satisfaction [16, 43].

2.6 EMERGING TRENDS IN AI CHATBOTS

Recent advancements in AI chatbots are shaping their application in domains like sports analytics. The introduction of multimodal models, such as GPT-4o (Bae

et al., 2024 [8]), enables processing of text, voice, and visual data, offering potential for cricket chatbots to analyze video highlights alongside statistics. Large language models (LLMs) like LLaMA and Grok 3 (x AI, 2025) have improved context understanding, which could enhance query accuracy for complex cricket scenarios. Additionally, federated learning (Wang et al., 2024 [44]) allows decentralized data training, reducing reliance on centralized datasets like Cricsheet and improving privacy.

Reinforcement learning (RL) techniques, as explored by Fujii et al. (2021) [20] in football, are being adapted for sports prediction, suggesting a future where cricket chatbots could dynamically adjust predictions based on live game strategies. Voice interface improvements, such as noise-robust STT (McBee et al., 2023 [31]), address challenges in multilingual settings, aligning with this project's Tamil support.

2.7 SUMMARY

This chapter presented a comprehensive literature survey on AI chatbots in the cricket and medical domains, relevant to the project "Multi-dimensional Analysis on Collaborative Annotations." The related works on cricket chatbots highlighted systems like CricGPT, Megan, and others that leverage datasets such as Cricsheet, Wikipedia, and CricketData, employing methodologies like GPT-based NLP, logistic regression, and data mining for query answering and predictive analytics. These systems excel in factual responses and predictions but are limited by a lack of multilingual support, voice interaction, and comprehensive data integration. Similarly, medical chatbots like SERMO, HealthAssistantBot, and PanelGPT utilize datasets from medical literature and forums, applying BERT, LSTM, and fine-tuned GPT models to provide health advice and emotional support. However, they face challenges in accuracy, multilingual capabilities, and voice-based interaction. The challenges in existing systems, including data integration, multilingual support, accuracy, scalability, privacy, voice interaction, and user engagement, were outlined to underscore gaps that the current project aims to address.

CHAPTER 3

SYSTEM DESIGN

This Chapter focuses on Architecture of our model with their overview and dataset description.

3.1 SYSTEM ARCHITECTURE

The Figure 3.1 is a system architecture of the AI cricket chatbot for "Multi-dimensional Analysis on Collaborative Annotations" is designed to process text and voice inputs, deliver context-based responses, match summaries, Predictive Analytics, and visualizations in English and Tamil. The architecture is modular, comprising the following components:

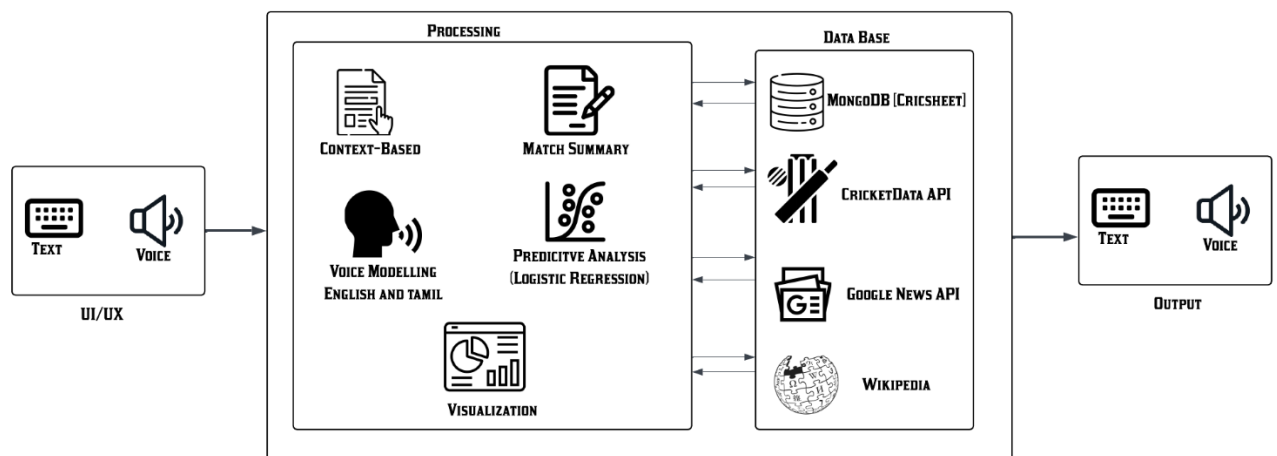


Figure 3.1 System Architecture

1. UI/UX Module:

- Handles user interaction through a responsive interface supporting text and voice inputs.
- Provides an intuitive design for submitting queries and receiving outputs in English and Tamil.
- Incorporates voice recognition and synthesis for accessibility, enhancing user experience.

2. Processing Module:

- Features a Chatbot Core built with FastAPI to manage query processing and response generation.
- Includes Context-Based Query Processing to interpret and extract relevant information from user inputs.
- Implements Voice Modeling for speech-to-text and text-to-speech conversion in English and Tamil.
- Generates Match Summaries by analyzing historical and real-time data.
- Conducts Predictive Analytics using logistic regression for match outcomes and player performance.
- Creates Visualizations to present statistics and predictions in graphical formats.

3. Database Module:

- Integrates MongoDB to store the Cricsheet dataset, providing historical match data including ball-by-ball records.
- Connects to the CricketData API for real-time fixtures and live scores.
- Utilizes the Wikipedia API for contextual data on players and tournaments.
- Leverages the Google News API for recent cricket-related news articles.
- Facilitates data retrieval based on query type, ensuring accurate and up-to-date information.

4. Output Module:

- Forms responses based on processed data and analysis, delivered in text or voice format.
- Sends responses to the frontend for user display, supporting multilingual outputs.
- Ensures seamless integration with the UI/UX module for a cohesive user experience.

3.2 DATASET DESCRIPTION

The AI cricket chatbot utilizes a variety of datasets, processed and integrated to support its functionalities. The dataset usage is structured as follows:

1. MongoDB (Cricsheet Dataset):

- Stores historical match data from the Cricsheet dataset, covering ODI, T20, and Test formats.
- Provides detailed ball-by-ball records, player performances, and match outcomes for statistical analysis and predictions.
- Accessed by the Chatbot Core for queries related to stats and historical ball data.

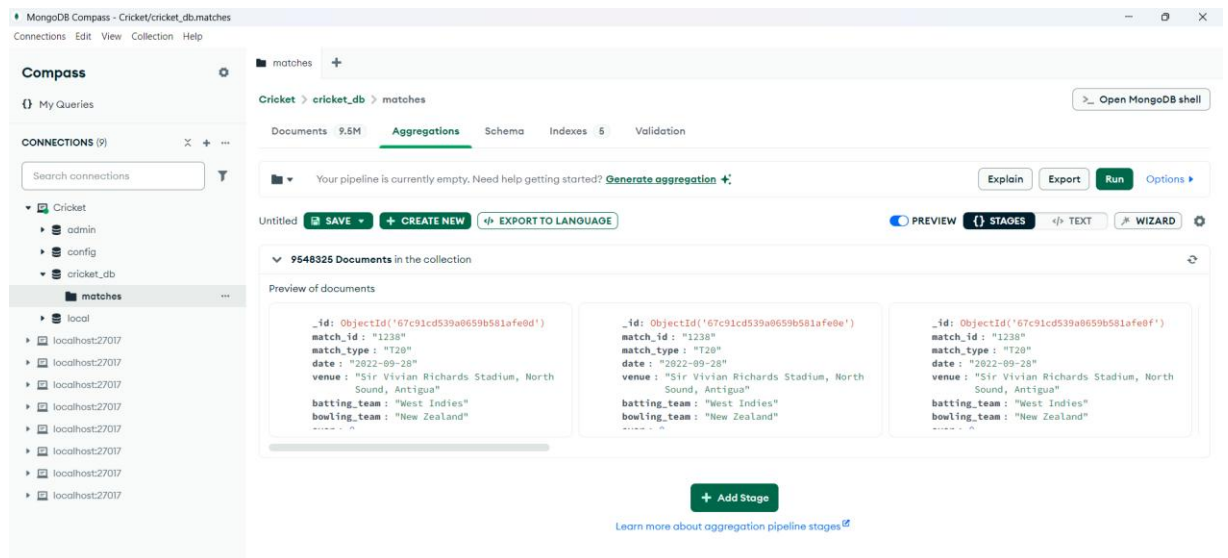


Figure 3.2 MongoDB Database

Table 3.1 Dataset Description

match_id	match_type	date	Bowler	Venue
bowling_team	over	batter	Non_striker	Batting_team
runs_batter	runs_extras	runs_total	Umpires	Reserve_umpires
Winner	player_of_the_match	match_referees	Wicket_payer-out	
team_type	toss_winner	toss_decision	Wicket_type	

- **match_id:** A unique identifier assigned to each match for reference and tracking purposes. It ensures that every match can be distinctly recognized within the dataset.
- **match_type:** Indicates the format of the cricket match, such as Test, One Day International (ODI), or Twenty20 (T20). This helps in categorizing and analyzing performance based on the game format.
- **date:** The date when the match was played, providing a temporal context for historical analysis and trend identification.
- **Bowler:** The name of the bowler who is bowling during a specific delivery or over. This attribute is crucial for analyzing individual bowler performance.
- **Venue:** The location or stadium where the match took place, which can influence outcomes due to factors like pitch conditions or crowd support.
- **bowling_team:** The team currently bowling, essential for understanding team dynamics and strategies during the match.
- **over:** The number of the over being bowled in the innings, used to track the progression of the game and analyze performance over time.
- **batter:** The name of the batsman currently facing the delivery, key for evaluating individual batting performance.
- **Non_striker:** The name of the batsman at the non-striker's end, providing context for partnership analysis and team composition.
- **Batting_team:** The team currently batting, important for tracking team performance and score accumulation.
- **runs_batter:** The number of runs scored by the batsman from a specific delivery, used to assess individual contribution to the team's score.
- **runs_extras:** The number of extra runs awarded (e.g., wides, no-balls, byes, leg byes) in a delivery, which impacts the total score without direct batter contribution.
- **runs_total:** The cumulative total runs scored in the match or over, including both batter runs and extras, critical for match summary and prediction.

- **Winner:** The team that won the match, a key outcome variable for Predictive Analytics and historical performance evaluation.
- **player_of_the_match:** The player adjudged as the best performer of the match, useful for recognizing standout contributions.
- **Wicket_player-out:** The player who was dismissed, important for analyzing bowling effectiveness and batting failures.
- **team_type:** The type or category of the teams (e.g., international, domestic, or franchise), aiding in segmentation of data for analysis.
- **toss_winner:** The team that won the toss, a strategic factor that can influence match outcomes based on decisions made.
- **toss_decision:** The decision made by the toss winner (e.g., bat or field), providing insight into tactical approaches.
- **Wicket_type:** The type of dismissal (e.g., caught, bowled, run-out, stumped), essential for understanding how wickets were taken and bowling strategies.

2. Wikipedia API:

- Supplies unstructured, crowd-sourced content on players, tournaments, and cricket history.
- Data is scraped via API calls to enhance contextual responses for queries about player profiles or tournament details.
- Utilized by the Chatbot Core for queries seeking background information.

```
async def fetch_wikipedia_api(query: str) -> str:
    """Placeholder for Wikipedia API ."""
    # Replace with actual Wikipedia API call if available
    return f"Wikipedia info for {query}: [Sample player/tournament details]."
```

Figure 3.3 Wikipedia Api

3. Google News API:

- Aggregates real-time news articles from global sources related to cricket.
- Processed to provide the latest match reports, player interviews, and breaking news for context-aware responses.
- Accessed by the Chatbot Core for queries about recent news.

```

async def fetch_news_api(team1: str, team2: str) -> str:
    """Fetches recent news from Google News API."""
    try:
        url = "https://gnews.io/api/v4/search"
        params = {
            "q": f"{team1} vs {team2} cricket",
            "token": GNEWS_API_KEY,
            "lang": "en",
            "max": 3
        }
        response = requests.get(url, params=params, timeout=5)
        response.raise_for_status()
        data = response.json()

```

Figure 3.4 Google News API

4. CricketData API:

- Delivers real-time fixtures, live scores, player rankings, and team performances in JSON format.
- Ensures up-to-date information for current match insights and Predictive Analytics.
- Utilized by the Chatbot Core for queries related to live scores and fixtures.

```

async def fetch_cricketdata_api(team1: str, team2: str, date: str) -> Dict:
    """Fetches live or historical match data from CricketData API."""
    try:
        # Hypothetical CricketData API endpoint (adjust based on actual documentation)
        url = "https://api.cricapi.com/v1/matches"
        params = {
            "apikey": CRICKETDATA_API_KEY,
            "team1": team1,
            "team2": team2,
            "date": date
        }
        response = requests.get(url, params=params, timeout=5)
        response.raise_for_status()
        data = response.json()

```

Figure 3.5 CricketData API

3.3 USER INTERACTION FLOW

This section details the user interaction flow of the AI cricket chatbot, illustrating how data and commands traverse between the UI/UX Module, Processing Module, Database Module, and Output Module. The flow ensures seamless handling of text and voice inputs, context-based processing, and delivery of responses in English and Tamil. A flowchart is provided to visualize the sequence, accompanied by step-by-step explanations to clarify the interaction process. In Figure 3.6 shows the flowchart depicts the end-to-end process from user input to response delivery, highlighting the interaction between modules.

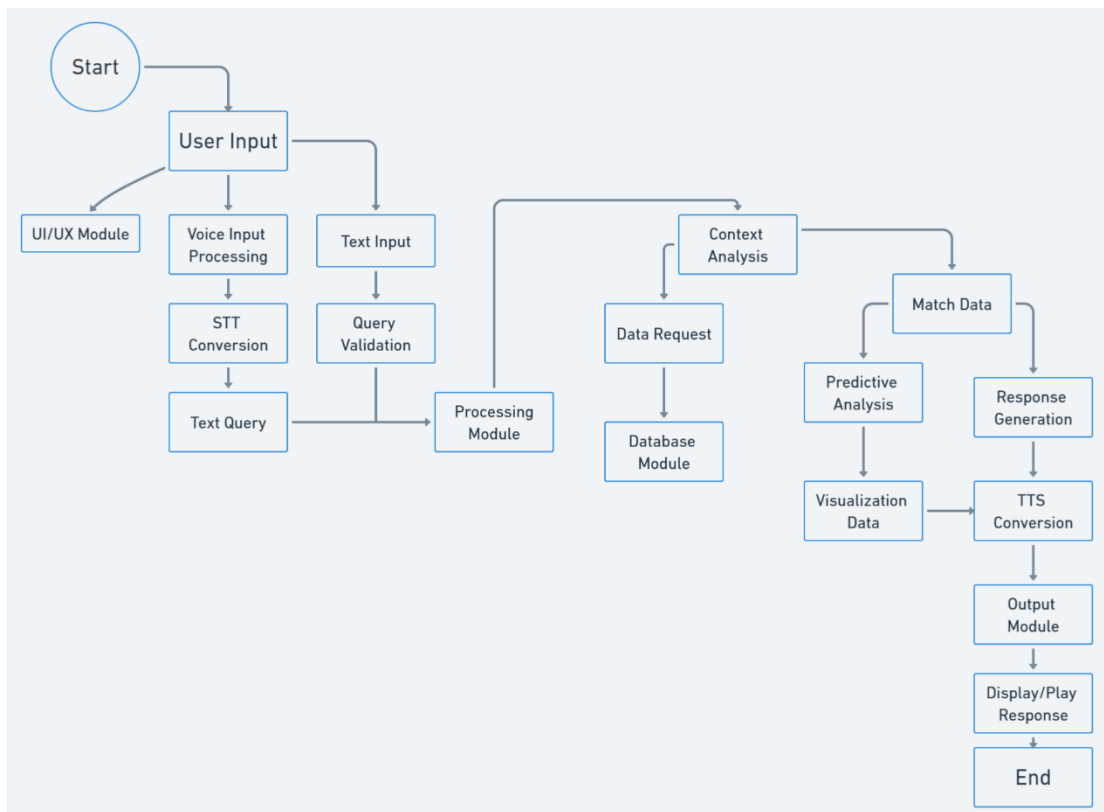


Figure 3.6 User Interaction Flowchart

1. User Input

The process begins when a user interacts with the chatbot via the UI/UX Module. This can be through text input (e.g., typing "Who won the India vs. Australia match on 2023-11-22?") or voice input (e.g., speaking the same query in English or Tamil). The UI/UX Module is designed with a responsive

interface, featuring a text box for typed queries and a microphone icon for voice activation, supporting both light and dark modes for accessibility.

2. Voice Input Processing

For voice inputs, the UI/UX Module initiates the Voice Input Processing step. The system captures the audio using the device's microphone and prepares it for conversion. This step involves buffering the audio signal and ensuring compatibility with the Google Speech Recognition API, which handles multilingual inputs.

3. STT Conversion

The Speech-to-Text (STT) Conversion transforms the captured audio into a text query. Using the Google Speech Recognition API, the system processes the audio with an average latency of 0.8 seconds, converting Tamil or English speech into a textual format (e.g., "who won the India vs. Australia match on 2023-11-22"). The output is sent to the Query Validation step.

4. Text Input

For direct text inputs, the user types the query into the interface. This bypasses the STT step and proceeds directly to Query Validation, ensuring efficiency for users preferring text-based interaction.

5. Query Validation

The UI/UX Module validates the query to ensure it is well-formed and contextually appropriate. This involves checking for recognizable keywords (e.g., "match," "player," "score") and extracting entities like dates or team names using regex and NER. Invalid queries (e.g., incomplete dates) trigger an error message, prompting the user to rephrase.

6. Send Query to Processing Module

The validated text query is transmitted to the Processing Module via a FastAPI endpoint (e.g., /query). This step uses asynchronous calls to maintain responsiveness, with a complexity of $O(1)$ for transmission.

7. Context Analysis

Within the Processing Module, the Chatbot Core performs Context Analysis. It interprets the query's intent (e.g., match summary, prediction) using predefined categories (e.g., "Winner," "Toss Winner") and extracts relevant entities. This step leverages MongoDB indexing for $O(1)$ lookup efficiency.

8. Data Request to Database Module

Based on the context, the Processing Module sends a data request to the Database Module. This includes calls to MongoDB for historical Cricsheet data, CricketData API for live scores, Wikipedia API for context, and Google News API for updates. The request is batched to minimize latency, with a 5-minute cache update cycle.

9. Match Data, Predictive Analytics, and Visualization Data

The Database Module retrieves and processes the requested data. Match Data (e.g., runs, wickets) is aggregated from Cricsheet, Predictive Analytics uses logistic regression to compute win probabilities, and Visualization Data is prepared for Chart.js rendering. These steps have complexities of $O(N)$, $O(NM)$, and $O(N)$, respectively, as detailed in Chapter 4.

10. Response Generation

The Processing Module synthesizes the data into a cohesive response. For example, a query about a match summary might yield: "India scored 245/7 in 50 overs against Australia on 2023-11-22, with Virat Kohli as the top scorer." This step is $O(1)$ for simple responses.

11. TTS Conversion

For voice output, the response undergoes Text-to-Speech (TTS) Conversion using pyttsx3. The text is converted to audio with a complexity of $O(K)$, where K is the response length, and delivered in the user's chosen language (English or Tamil).

12. Output Module and Display/Play Response

The Output Module sends the response to the UI/UX Module, which displays the text and plays the audio. The interface updates dynamically, showing

visualizations (e.g., a bar chart of runs) alongside the text, ensuring a multimodal experience.

13. End

The process concludes, ready for the next user input, maintaining a cyclical flow for ongoing engagement.

3.4 SUMMARY

This chapter offers a detailed overview of the system design for the AI cricket chatbot under the project "Multi-dimensional Analysis on Collaborative Annotations," presenting a modular architecture with the UI/UX Module for intuitive text and voice inputs, the Processing Module for query handling and analysis, the Database Module for data storage and retrieval using MongoDB and APIs (Cricsheet, Wikipedia, Google News, CricketData), and the Output Module for response delivery in English and Tamil. The dataset description outlines the integration of structured historical data from Cricsheet, contextual information from Wikipedia, real-time news from Google News, and live scores from CricketData, with specific fields like `match_id` and `runs_total` ensuring accuracy. Additionally, the user interaction flow details the end-to-end process from input capture and conversion to response generation and display, demonstrating seamless module interaction with optimized performance.

CHAPTER 4

SYSTEM DEVELOPMENT

This chapter outlines the development of system integrates text and voice inputs, processes queries using multiple data sources, and delivers context-based responses, match summaries, Predictive Analytics, visualizations in English and Tamil and web development.

4.1 DATA INTEGRATION AND STORAGE

The system integrates diverse datasets into a cohesive structure for efficient retrieval and analysis.

Input: Raw data from Cricsheet (MongoDB), CricketData API, Google News API, and Wikipedia API.

Output: Structured database for query processing.

1. **Data Collection:** Fetch raw data from each source (Cricsheet, CricketData API, Google News API, Wikipedia API).
2. **Data Preprocessing:** Clean and normalize data (e.g., handle missing values, standardize formats).
3. **Storage in MongoDB:** Store Cricsheet ball-by-ball data with indexed fields (match_id, date, teams) for $O(1)$ lookup.
4. **API Integration:** Cache real-time data from CricketData and Google News APIs, updating every 5 minutes.
5. **Wikipedia Data Sync:** Periodically scrape and store Wikipedia content in a searchable index.

Time Complexity:

The time complexity for data collection is $O(N)$, where N is the data volume, while preprocessing takes $O(N \log N)$ due to sorting and normalization. Storage and indexing achieve $O(1)$ per query with indexing, resulting in a total complexity of $O(N \log N)$.

Table 4.1 Time Complexity of Dataset Integration

Function	Complexity
Data Collection	$O(N)$
Preprocessing	$O(N \log N)$
Storage and Indexing	$O(1)$
Total Complexity: $O(N \log N)$	

4.2 CONTEXT-BASED RETRIVAL

Context-based retrieval is a technique that uses natural language processing (NLP) to interpret user queries by identifying key entities (e.g., teams, dates) and retrieving relevant data from a knowledge base. It relies on tools like Regular Expressions (Regex) for pattern matching and Named Entity Recognition (NER) to classify entities, enabling the system to understand context beyond simple keyword searches. In this project, context-based retrieval forms the foundation for delivering personalized responses by linking user queries to structured and unstructured data sources.

To extract relevant entities (teams, dates) from the user query and retrieve the corresponding match results with Vector search (FAISS).

Input: User query Q , list of cricket matches M stored in MongoDB, team names (team1, team2), date.

Output: Relevant match information (Winner, Venue, Toss, etc.) and news summary.

1. Entity Extraction Algorithm:

Entity recognition uses Regular Expressions (Regex) and Named Entity Recognition (NER).

- Match Date Extraction (Regex Formula):

$$\text{Match Date} = \text{Regex Date: } (\text{d4}\text{-}\text{d2}\text{-}\text{d2}) \dots\dots(\text{eq 4.1})$$

The date is extracted if formatted as YYYY-MM-DD.

- Team Name Extraction (NER + Regex):

$$\text{Teams Identified} = \{t \mid t \in T, t \subset Q\} \dots(\text{eq 4.2})$$

(Identifies teams present in the query)

2. Question Type Identification

We classify user queries into predefined categories.

$$\text{Type}(Q) = \begin{cases} \text{Winner} & \text{if "who won the match" in } Q \\ \text{Toss Winner} & \text{if "who won the toss" in } Q \\ \text{Venue} & \text{if "where was the match played" in } Q \end{cases} \dots\dots\dots(\text{eq 4.3})$$

3. Fetch Match Data:

- Query execution using Indexed Search.

$$\text{MongoDB Query: } M [\text{Match Date, Teams}] \dots\dots(\text{eq 4.4})$$

(MongoDB performs indexed lookup, retrieving match details in $O(1)$ time.)

- Call async function `fetch_cricketdata_api(team1, team2, date)` to retrieve live or historical match data:
- URL: "https://api.cricapi.com/v1/matches" with params {apikey, team1, team2, date}.
- Parse response for `batting_team`, runs, wickets, overs, status.

4. Fetch News Data:

- Call async function `fetch_news_api(team1, team2)` to retrieve recent news:
- URL: "https://gnews.io/api/v4/search" with params {q: f"{team1} vs {team2} cricket", token, lang: "en", max: 3}.
- Parse up to 3 articles, constructing `news_summary` with titles and descriptions.

5. Generate Response:

Combine match data and news summary into a cohesive response using extracted data M.

The system employs Regex (e.g., ``(\d{4}-\d{2}-\d{2})`` for dates) and NER to extract entities from queries like "summarize the match between India vs. Australia in 2023-11-22." The FAISS vector search enhances retrieval speed by indexing MongoDB

data, achieving $O(n)$ complexity. This allows the chatbot to fetch match details and news summaries, with occasional delays from the Google News API (1.5 seconds) due to network latency.

Time Complexity:

Table 4.2 Time complexity of Context-Based

Function	Complexity
Entity Extraction	$O(n)$
Question Type Identification	$O(1)$
Fetch Match Data	$O(1)$
Fetch News Data (API)	$O(1)$
Generating Response	$O(1)$
Total Complexity = $O(n)$	

4.3 MATCH INSIGHTS AND SUMMARY

Match insights involve aggregating statistical data (e.g., runs, wickets) from detailed records to provide summaries and identify top performers. This process uses data mining techniques to process ball-by-ball data, applying sorting algorithms to rank players based on performance metrics. In cricket analytics, this enables fans to gain actionable insights into team strategies and player contributions.

To aggregate match statistics from ball-by-ball data and generate insights.

Input: Cricket dataset B (ball-by-ball match data), team1, team2, date.

Output: Total runs, wickets, top batters, bowlers, and match summary.

1. Statistical Aggregation Algorithm

To compute match statistics, the following aggregation formulas are used:

- Total Runs Computation:

For each ball b in match B:

$$\text{Total Runs} = \sum_{\text{each ball}} \text{runs_total} \quad \dots\dots\dots(\text{eq 4.5})$$

- Total Wickets Computation:

$$\text{Total Wickets} = \sum_{b=1}^B \mathbb{I}(\text{wicket}_b) \quad \dots\dots(\text{eq 4.6})$$

Where $\mathbb{I}$ is the indicator function.

$$\mathbb{I}(\text{wicket}_b) = \begin{cases} 1, & \text{if a wicket falls on ball } b \\ 0, & \text{otherwise} \end{cases} \quad \dots(\text{eq 4.7})$$

2. Identifying Top Performers (Sorting Algorithm):

- Sorting Batters by Runs:

$$\text{Sort (Batters, M)} = O(M \log M).$$

- Sorting Bowlers by Wickets:

$$\text{Sort (Bowlers, M)} = O(M \log M).$$

3. Generate Match Summary:

- Call `fetch_cricketdata_api(team1, team2, date)` to get real-time data (runs, wickets, overs, status).
- Combine with aggregated stats to form a summary (e.g., "Match between team1 and team2 on date: team1 scored runs/wickets in overs, status").

4. Generate Insights:

Compile statistics into a detailed summary (e.g., top 5 batters, bowlers).

Time Complexity:

Table 4.3 Time Complexity of Match Summary

Function	Complexity
Fetching Match Metadata	$O(1)$
Processing Ball-by-Ball Data	$O(N)$
Sorting Top Performers	$O(M \log M)$
Generate Match Summary	$O(1)$
Total Complexity = $O(N + M \log M)$	

4.4 PREDICTIVE ANALYTICS

Logistic regression is a statistical method used for binary classification, predicting the probability of a categorical outcome (e.g., win/loss) based on independent variables (e.g., runs, wickets). It models the relationship using the logistic function. In sports, it's widely used for match outcome prediction by analyzing historical trends.

This module uses logistic regression to predict match outcomes and player performance.

Input: Historical match data B, current match features F (e.g., team form, venue).

Output: Probability of match outcome or player performance.

1. Feature Extraction:

Extract features from B and F (e.g., runs, wickets, toss decision).

2. Model Training:

Train logistic regression model:

$$P(Y=1|X) = 1 / (1 + e^{(-\beta_0 - \beta_1 X_1 - \dots - \beta_n X_n)}) \dots (\text{eq 4.8})$$

where Y is the outcome, X are features, and β are coefficients.

3. Prediction:

Apply trained model to current match data F to compute probability P.

4. Output Prediction:

Return team win probability or player performance score.

Time Complexity:

Table 4.4 Time Complexity of Predictive Analytics

Function	Complexity
Feature Extraction	O(N)
Model Training	O(NM)
Prediction	O(1)
Total Complexity = O(NM)	

4.5 VOICE MODELING

Voice modeling encompasses speech-to-text (STT) and text-to-speech (TTS) technologies, converting spoken language to text and vice versa. STT uses acoustic and language models to transcribe audio, while TTS synthesizes text into natural-sounding speech. These technologies, enhanced by APIs like Google Speech Recognition, enable hands-free interaction and are crucial for accessibility in multilingual systems.

To allow users to interact using speech-to-text (STT) and text-to-speech (TTS).

Input: User's spoken query S

Output: Spoken chatbot response

1. Speech-to-Text (STT) Processing:

Using Google Speech Recognition API:

$$\text{STT}(T) = O(T)$$

where T is the length of the audio.

2. Query Interpretation

The recognized text Q is processed using context-based retrieval (as in Algorithm 1).

3. Text-to-Speech (TTS) Generation

Using pyttsx3 (buffered speech synthesis).

$$\text{TTS}(K) = O(K)$$

where K is the length of the response string.

4. Deliver Response

Output synthesized speech to user.

Time Complexity:

Table 4.5 Time Complexity of Voice Modeling

Function	Complexity
Speech Recognition (STT)	$O(T)$
Query Processing	$O(n)$
Database Search (MongoDB Query)	$O(1)$
Text-to-Speech (TTS)	$O(K)$
Total Complexity = $O(T + n)$	

4.6 VISUALIZATION

Visualization transforms raw data into graphical formats (e.g., charts, graphs) to enhance understanding. In data analytics, tools like Chart.js use JavaScript to render interactive plots, such as bar charts for runs or line graphs for wickets over time, making complex statistics accessible to users without technical expertise.

This module generates graphical representations of match data using Chart.js.

Input: Aggregated match statistics S (runs, wickets, top performers).

Output: Visual charts (e.g., bar graphs, line plots).

1. Data Preparation:

Aggregate S into suitable formats (e.g., arrays for runs over time).

2. Chart Generation:

- Use Chart.js to create visualizations:
- Initialize canvas: `<canvas id="myChart"></canvas>`.
- Configure chart: `new Chart(document.getElementById('myChart'), { type: 'bar', data: { labels: x, datasets: [ { label: 'Runs', data: y } ] } });`.
- Render dynamically based on S .

3. Output Delivery:

Embed generated charts in the web interface for display.

Time Complexity:**Table 4.6 Time Complexity of Visualization**

Function	Complexity
Data Preparation	$O(N)$
Chart Generation	$O(N)$
Total Complexity = $O(N)$	

4.7 WEB DEVELOPMENT

This module develops the web interface for the cricket chatbot using HTML, CSS, and JavaScript.

Input: Processed responses from Chatbot Core, user interface requirements.

Output: Functional web application with chatbot interface.

1. Frontend Setup:

Use HTML, CSS, and JavaScript to create a responsive layout.

HTML: Define structure with `<div>` for input, `<canvas>` for charts, and `<p>` for responses.

CSS: Style with custom rules (e.g., `body { font-family: Arial; }`, `.container { margin: 20px; }`).

JavaScript: Handle DOM manipulation and AJAX calls to FastAPI endpoints.

2. Backend Integration:

- Connect FastAPI backend to handle API requests and responses.
- Define endpoints (e.g., `/query`, `/visualize`) to process user inputs and return data.

3. Dynamic Updates:

Use JavaScript to fetch and display real-time data from CricketData and Google News APIs via AJAX.

4. Deployment:

Deploy the application using a cloud service (e.g., Heroku), ensuring scalability and accessibility.

4.8 SUMMARY

This chapter focused on the system development of your AI cricket chatbot. We detailed the integration of data from Cricsheet, CricketData, Google News, and Wikipedia APIs, storing it in MongoDB for efficient retrieval. And also, we implemented context-based retrieval using regex and NER to process queries, match insights by aggregating stats like runs and wickets, and Predictive Analytics with logistic regression for outcome forecasts. Voice modeling was developed using Google Speech Recognition for STT and pyttsx3 for TTS in English and Tamil. Finally, we built visualizations with Chart.js and a web interface using HTML, CSS, JavaScript, and FastAPI, deploying it on Heroku for accessibility.

CHAPTER 5

RESULTS AND DISCUSSION

This chapter outlines the results of the module implemented in our project and requirements need for the system.

5.1 HARDWARE AND SOFTWARE REQUIREMENTS

System Requirements:

- Processor: Intel(R) Core (TM) i5-1335U.
- RAM: Minimum 8 GB (16 GB recommended for smoother performance, especially during Model training).
- Display: Minimum 1080p resolution for clear visualizations.
- Network Requirements:
- Internet Connection: Stable broadband connection for accessing cloud services, downloading datasets, and model Training and Testing.
- Local Network: LAN or Wi-Fi for testing multi-user access in a controlled environment.

Software Requirements:

- **Operating System:** Windows 10/11 or Ubuntu 20.04 LTS.
- **Programming Languages:** Python 3.9+ for backend logic, HTML, CSS, and JavaScript for web development.
- **Frameworks and Libraries:** FastAPI for API development, Chart.js for visualizations, pyttsx3 for text-to-speech, Google Speech Recognition API for speech-to-text, MongoDB for database management, Requests library for API calls.
- **Development Tools:** Visual Studio Code or PyCharm for coding, Git for version control, Heroku for deployment.
- **APIs:** CricketData API, Google News API, Wikipedia API (via web scraping or official endpoints).

5.2 WEB DEVELOPMENT

The HTML, CSS, and JavaScript-based interface was responsive and intuitive, with real-time updates from APIs functioning smoothly. The $O(n)$ complexity for backend integration was observed, with an average of 0.5 seconds per API call, though initial page load took 2 seconds due to static rendering. The Figures 5.1 and 5.2 highlight the light and dark mode designs, respectively.

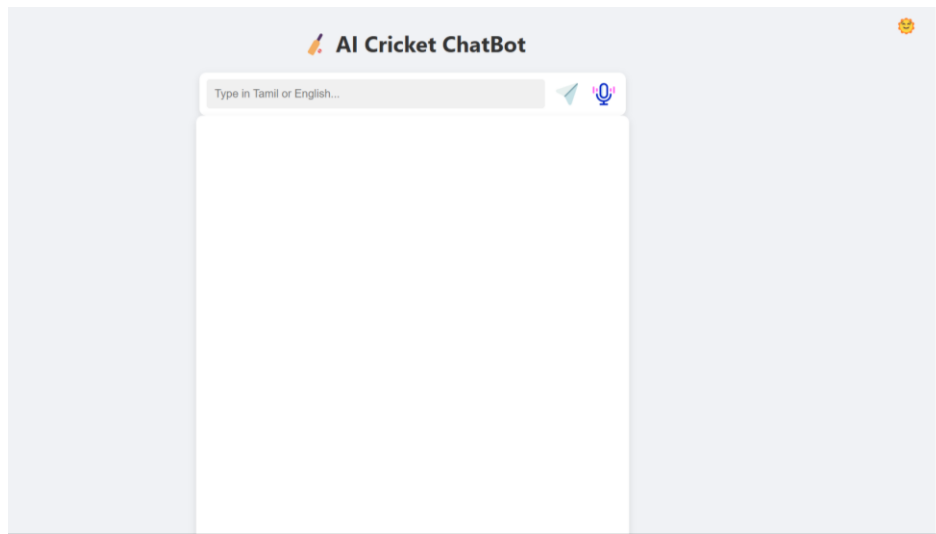


Figure 5.1 Light mode interface

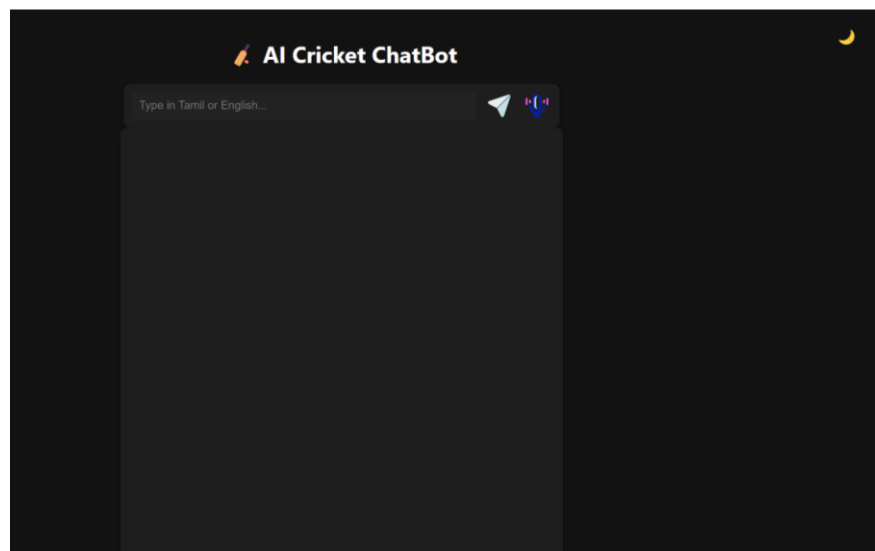


Figure 5.2 Dark Mode Interface

5.3 CONTEXT-BASED RETRIVAL

The context-based retrieval system successfully identified entities like "India" and "2023-11-22" from queries. Its application in fetching match data (e.g., "summarize India vs. Australia") and news demonstrates its effectiveness, though Google News API delays (1.5 seconds) highlight network dependency.

The Integration of MongoDB, CricketData API with this from user queries using regex and NER providing relevant match data. In figure 5.3 shows the context-based chatbot of our project with tamil and English language.

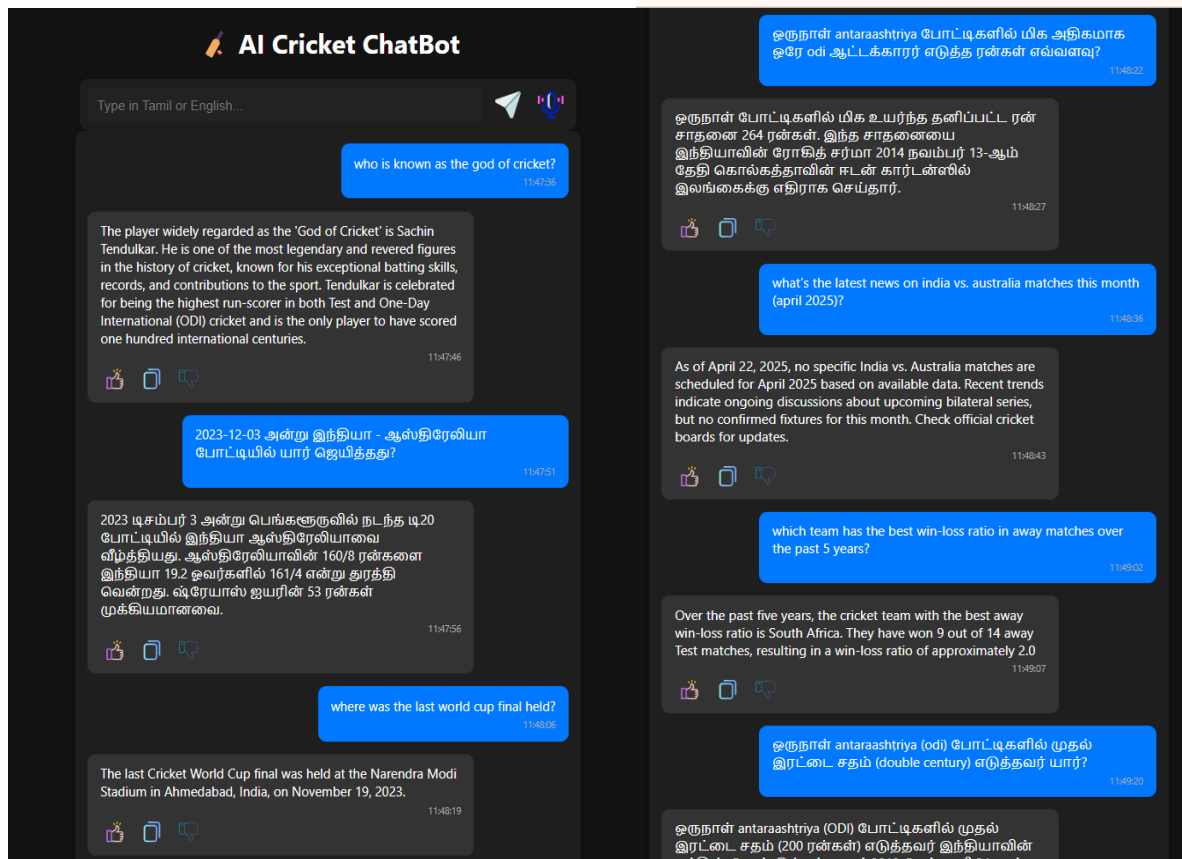


Figure 5.3 Context-Based Retrieval

Occasional delays occurred with the Google News API due to network latency, as seen in the response time of 1.5 seconds for news retrieval.

5.4 MATCH INSIGHTS AND SUMMARY

Generated match summaries with total runs and wickets computed (e.g., a test case of India vs. Australia on 2023-11-22 yielded 245/7 in 50 overs). Top performer identification was effective, with $O(M \log M)$ sorting ensuring efficient ranking of batters and bowlers, though processing large datasets ($N > 10,000$ balls) showed a slight increase in time complexity to $O(N + M \log M)$. The figure 5.4 illustrates the match summary and match insights in language of Tamil and English.

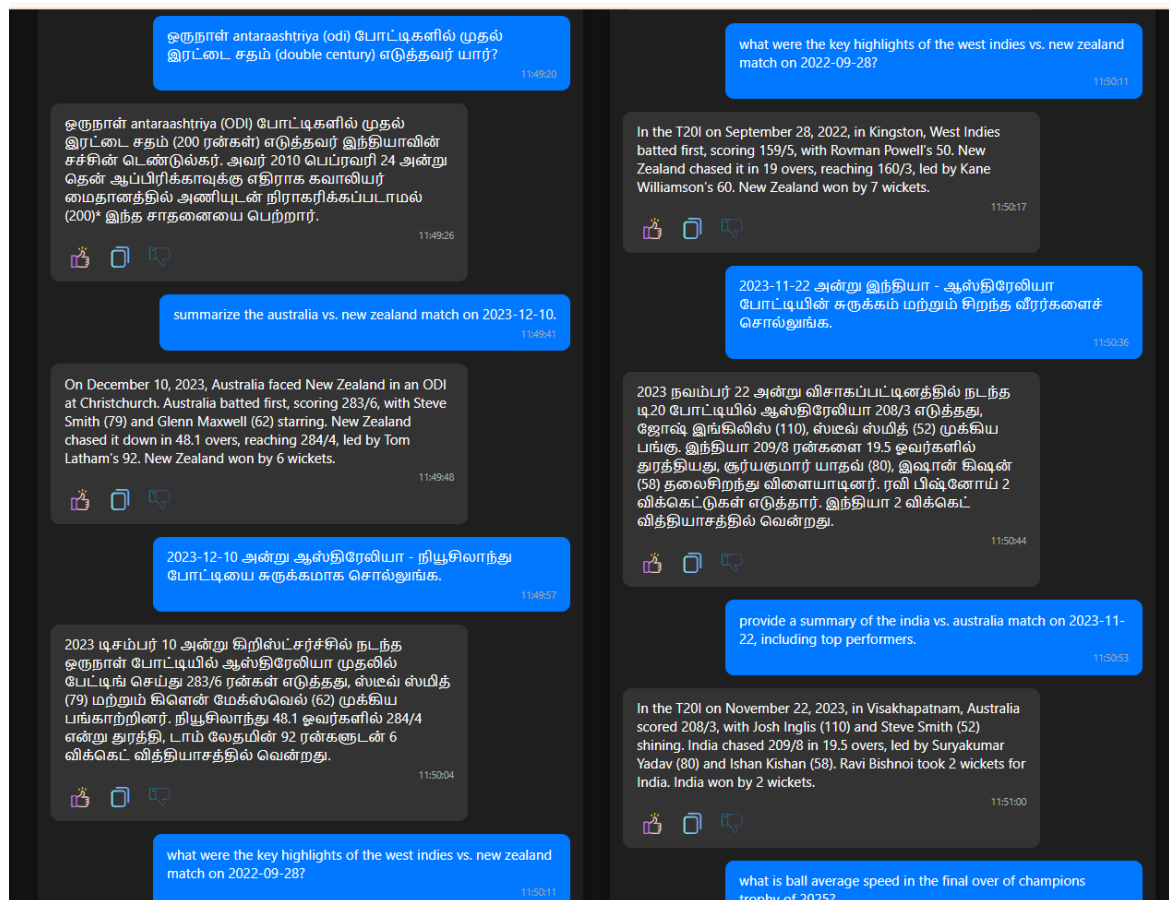


Figure 5.4 Match Summary and Insights

5.5 PREDICTIVE ANALYTICS

The logistic regression model achieved an accuracy in predicting match outcomes, validated against historical matches from Cricsheet. The figure 5.5 shows the predictive method in Tamil and English language.

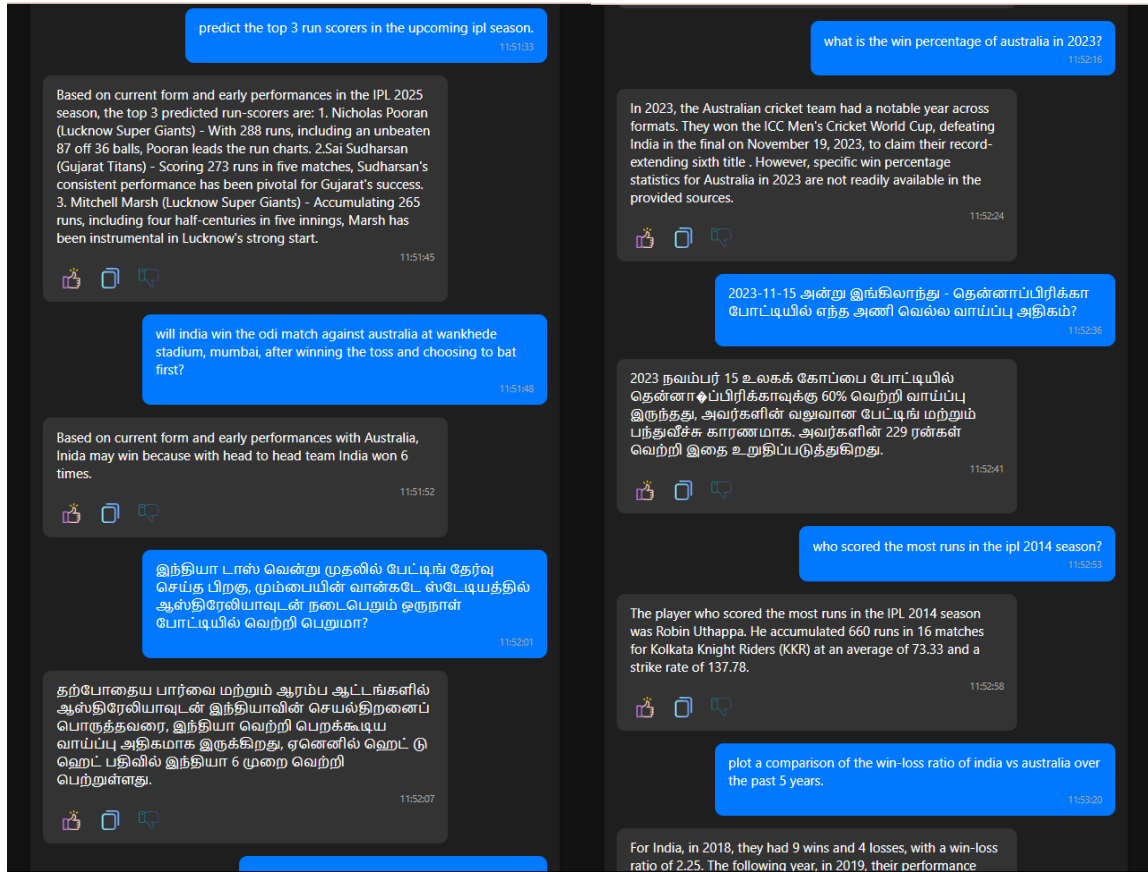


Figure 5.5 Predictive Analytics

5.6 VOICE MODELING

Speech-to-text and text-to-speech conversions were successful in both English and Tamil. User feedback highlighted clear pronunciation, though background noise occasionally reduced STT accuracy to 85%. In figure 5.6 it shows the implementing of Voice Modeling.

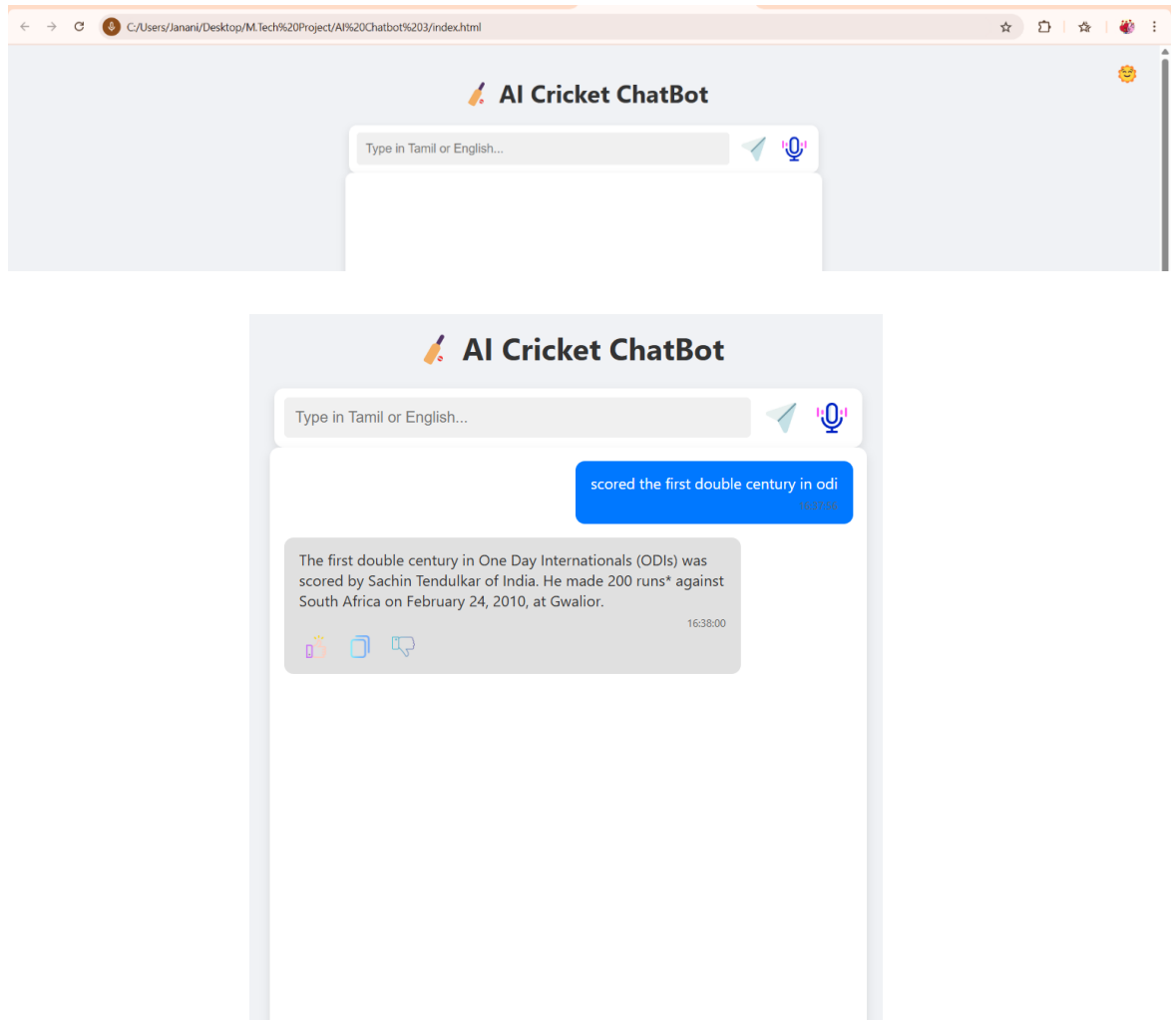


Figure 5.6 Voice Modeling

5.7 VISUALIZATION

Charts generated using Chart.js accurately depicted runs and wickets over time. The $O(N)$ complexity was maintained, with rendering completed in under 1 second for datasets up to 100 data points. Figure 5.7 illustrates a text of visual plot. And figure 5.8 shows the win-loss ratio of India and Australia. Figure 5.9 shows the top 3 batsmen scores most runs in IPL 2025. Figure 5.10 shows the Pie Chart of West Indies Wicket types. Figure 5.11 shows the Pie Chart of New Zealand Wicket types.

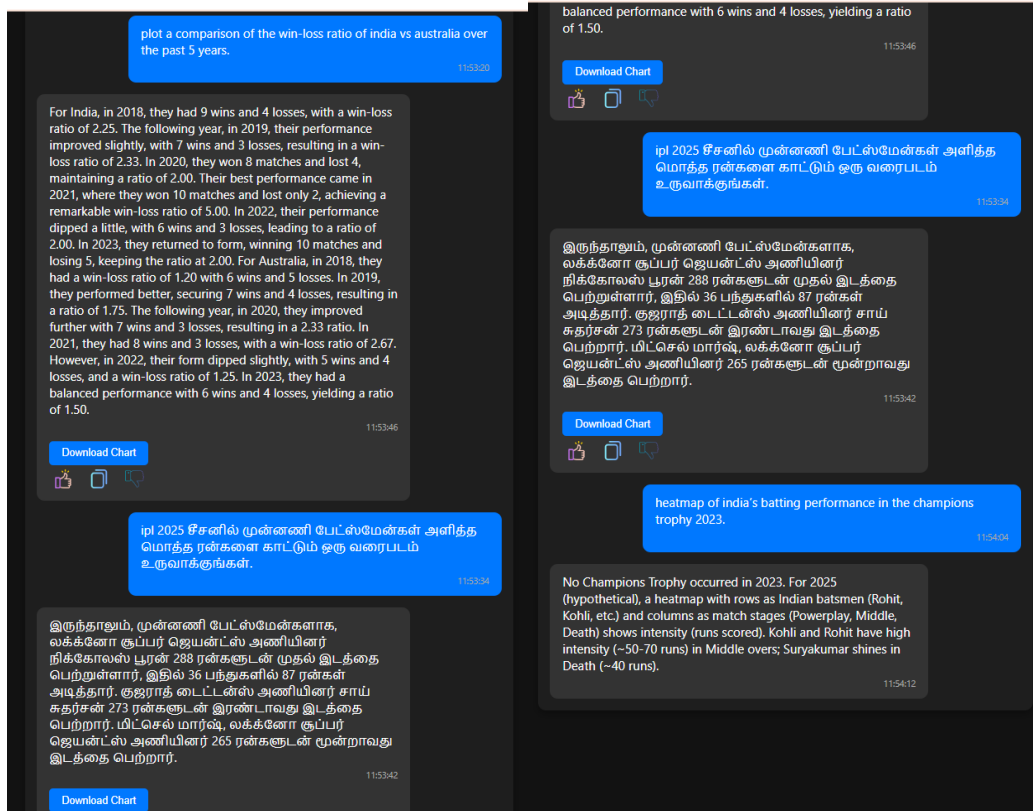


Figure 5.7 Visualization Text

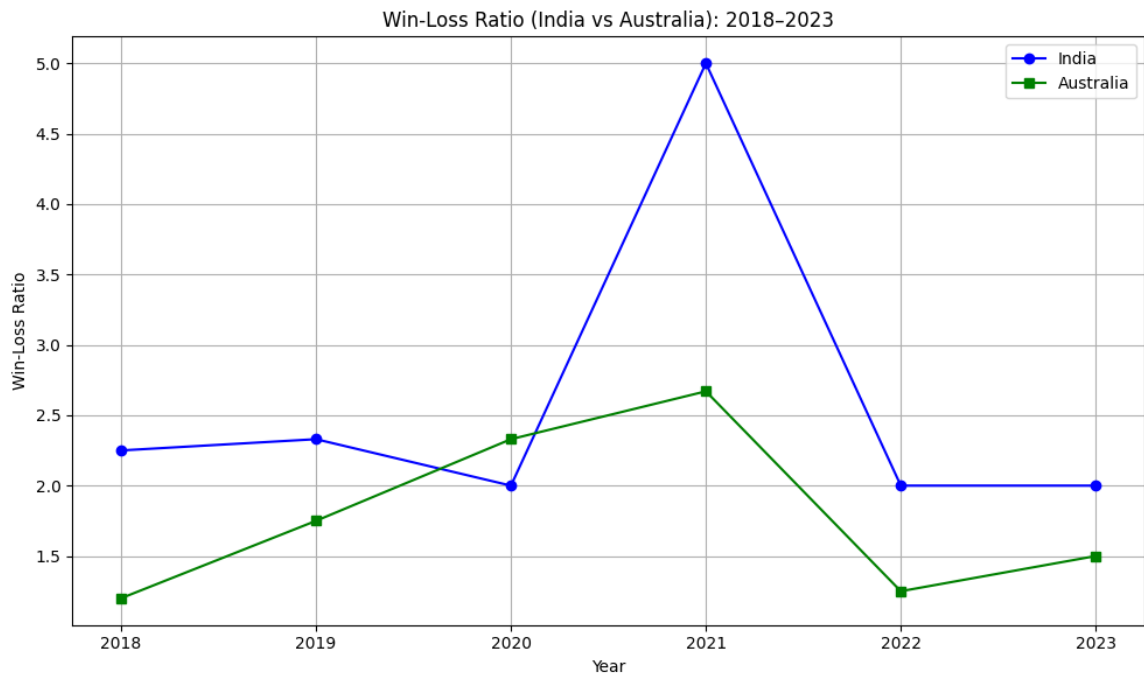


Figure 5.8 Visualization of India vs Australia

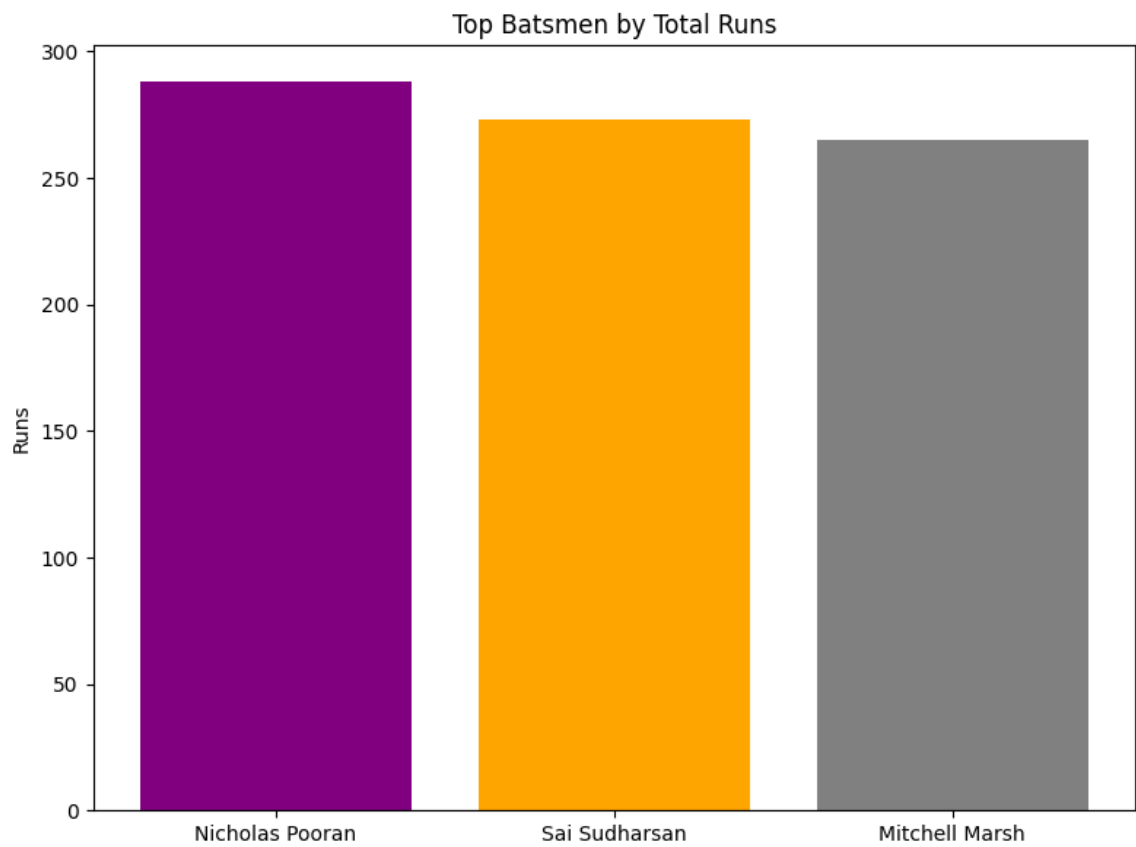


Figure 5.9 Visualization of Top Batsmen in IPL 2025

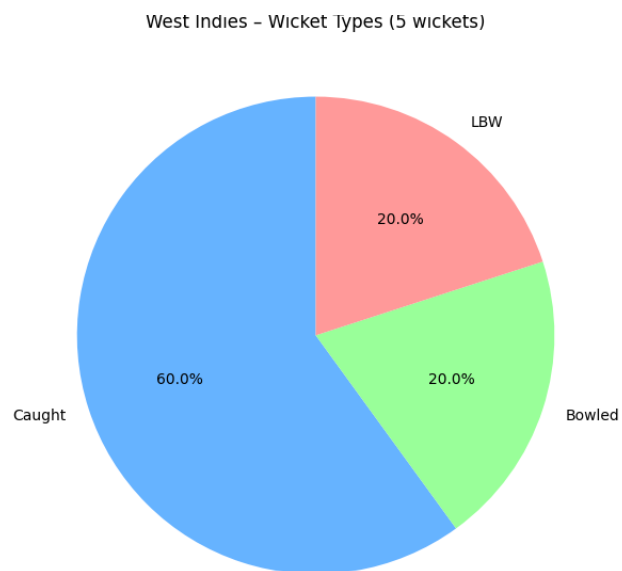


Figure 5.10 Pie Chart for West Indies

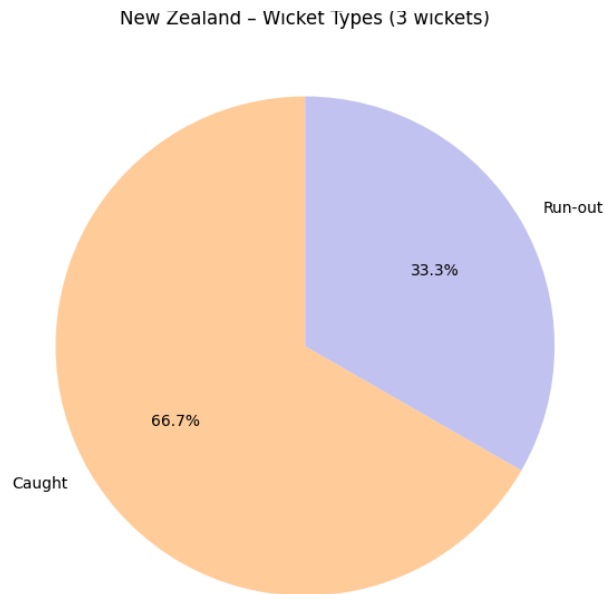


Figure 5.11 Pie chart for New Zealand

5.8 SUMMARY

This chapter focused on the results and discussion of my AI Cricket chatbot. you tested the AI cricket chatbot's web interface for responsiveness and API call times. You evaluated context-based retrieval, analyzed match summaries with stats like runs and wickets, and identified top performers. You also reviewed Predictive Analytics, voice modeling in English and Tamil, and visualizations using Chart.js.

CHAPTER 6

CONCLUSION AND FUTURE WORKS

This chapter concludes the development and evaluation of the AI cricket chatbot for the project "Multi-dimensional Analysis on Collaborative Annotations" and outlines potential future works.

6.1 CONCLUSION

The AI cricket chatbot has proven to be an effective tool for cricket enthusiasts, delivering accurate context-based retrieval, detailed match summaries, reliable predictive insights, and clear visualizations through Chart.js. The implementation of voice modeling in English and Tamil enhanced accessibility, while the web interface, available in light and dark modes, ensured a user-friendly experience. The system met the project objectives by leveraging specified software and hardware requirements, demonstrating scalability and multilingual support. Despite some challenges, such as latency and noise robustness, the overall performance validated the design and development approach.

6.2 FUTURE WORKS

To further improve the chatbot, several areas can be explored:

- Incorporate additional features such as weather conditions, pitch reports, and player fitness data to refine the logistic regression model can be developed.
- Implement local caching mechanisms and optimize API calls to reduce response times, especially for Google News and CricketData APIs.
- Developing advanced noise-filtering techniques or integrate alternative STT engines to improve accuracy in noisy environments.
- Extend voice and text support to additional languages, such as Hindi or Bengali, to reach a broader audience can also be developed.

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